

Honest-State Ledger v0.11 — Monday 2026-06-01 substantive program inflection. Cal #35 STANDING ratified (6-element discipline stack closes). 4 Casey-named principle candidates NAMED. P1 v0.7 SUBMISSION-FINAL package handed off (Casey dispatch pending). Matrix element framework framework-complete Steps 1-5. K206 audit framework filed (7 gates). Tier 0 v0.2 consolidation Session 2 triggered. 16 cumulative audit-chain events.

Keeper (Monday 2026-06-01)

Honest-State Ledger v0.11 — Monday 2026-06-01

0. What changed Monday — program-level inflection

Four convergent events make Monday a substantive program inflection:

- 1. Calibration #35 STANDING RATIFIED — six-element discipline stack closes
- 2. 4 Casey-named principle CANDIDATES NAMED — Casey approved names
- 3. P1 v0.7 SUBMISSION-FINAL package handed off — Casey dispatch action pending
- 4. Matrix element framework framework-complete Steps 1-5 — G chain at closed-form structure
- 5. Tier 0 v0.2 consolidation Session 2 triggered — Lyra + Keeper joint multi-hour work active
- 6. K206 audit framework filed — 7 explicit gates for matrix-element verification
- 7. Mon-event-4 logged — Elie 0.924 → 0.462 arithmetic walk-back caught by Keeper numerical reconstruction
- 8. Cumulative audit-chain 16 events (Sat 7 + Sun 5 + Mon 4)

1. Casey-named principles disposition

#	Principle	Status
1	Substrate Working Process Principle (SWPP)	STANDING
2	Five-Absence Predictions Set	STANDING
3	Substrate Closure Principle	STANDING
4	Graph Forces Principle	STANDING
5	Integer Web Principle	STANDING
6	Substrate Cognition Network Hypothesis	STANDING
7	D_IV ⁵ Rigidity	STANDING
8	SCMP	STANDING
9	(open slot)	—
10	(open slot)	—

#	Principle	Status
11	“Energy needs an anchor”	CANDIDATE (Saturday filed)
12	“Substrate Bulk-Boundary Projection”	NAMED CANDIDATE (Monday)
13	“Per-Generation Cluster Independence”	NAMED CANDIDATE (Monday)
14	“Substrate-Selected 4D Dimensionality”	NAMED CANDIDATE (Monday)
15	“Gravity is Light’s Momentum Shifted by Substrate”	NAMED CANDIDATE (Monday)

Casey signal: “ok, you can use the names” — approves use of names. NOT promotion to STANDING.

Honest tier: NAMED CANDIDATES. Promotion to STANDING requires Cal cold-read + K-audit ratification per Elie’s tier note.

Cal #35 audit pending: Elie flagged “effective independent principles (post Cal #35 unification): 2-3 depending on Candidate 3 scope decision.” Candidate 14 may be facet of Candidate 12 per Toy 3683 unification. Casey scope decision deferred.

2. Calibration #35 STANDING RATIFIED

Casey signal: All four signals GO — Cal #35 STANDING ratification APPROVED.

Six-element discipline stack CLOSED: 1. Cal #27 STANDING (Coincidence-filter at peak coherence) 2. Cal #29 STANDING (Mode-6 enumeration protocol) 3. Cal #32 STANDING (Parameter-role verification = integrality test) 4. Calibration #33 STANDING (RECALLED vs VERIFIED-CITED) 5. Calibration #34 STANDING (Narrow framing / conditional-tag-with-headline) 6. Calibration #35 STANDING (Independence-taxonomy-before-multiplicative-null-model) — NEW

Methodology Index v0.11 Q17 integration: Cal files formal entry.

Ratification basis (7 firings of operational evidence): - Sat: $g + \text{rank} = N_c^2$ 3-context independence reuse (3 firings: Weinberg + bulk-color + $V_{(1,1)}$ decomp) - Sun: m_W/m_Z arithmetic equivalence to P1 §7 existing claim (Keeper walk-back) - Mon-1: factor 2 double-promotion catch (Keeper-on-Grace+Lyra) - Mon-2: diagonal/off-diagonal parallel-framework catch (Cal pre-stage) - Mon-3: Pochhammer-+1-anomaly mechanism over-promotion (Keeper-on-Elie)

Cal #35 honest framing now operationally STANDING: prevents multiplicative null-model misuse of substrate-coincidence.

3. Paper P1 dispatch status

Lyra package handed off: - notes/Lyra_Paper_Substrate_Hall_Algebra_Primary_Draft_v0_2.pdf (54.9 KB) — main paper PDF v0.7 SUBMISSION-FINAL - notes/Lyra_Paper_P1_Finalize_Submission_Package.pdf (32.9 KB) — Section 2 cover letter (dated June 1, 2026) + Section 4 supplementary materials list

Casey dispatch action pending: - (a) Email Advances in Mathematics editor with cover letter + PDF + supplementary - (b) Concurrent arXiv preprint (math.RT + math.QA categories) - (c) Zenodo archive on publication

Engagement test status: ACTIVE upon dispatch. Cal’s “outside voice” question (Saturday) becomes empirically answerable on referee-cycle timescale.

4. Matrix element framework status — framework-complete Steps 1-5

G chain reduced form (Elie Toy 3692 reconciled):

$$G_{\text{predicted}} \propto (4\sqrt{2} \cdot c_{\text{FK}} / (n_{\text{C}} \cdot \sqrt{C_2} \cdot \hbar_{\text{BST}})) \cdot \ell_{\text{B}} \cdot \text{dim_bridge}$$

$$\approx 0.603 \cdot \ell_{\text{B}} / \hbar_{\text{BST}} \cdot \text{dim_bridge} \text{ (with } c_{\text{FK}} \text{ explicit)}$$

$$\text{OR } \approx 0.462 \cdot \ell_{\text{B}} / \hbar_{\text{BST}} \cdot \text{dim_bridge} \text{ (} c_{\text{FK}} \text{ absorbed into } \ell_{\text{B}} \text{ normalization)}$$

Mon-event-4 absorbed: Elie's "0.924" was rationalization arithmetic typo ($4\sqrt{2}/(5\sqrt{6}) \neq 8/(5\sqrt{3})$). Toy 3692 reconciled with correct value. Honest walk-back.

All factors substrate-clean and pinned: - $\Delta C_2 = 2$ at B_2 substrate (Heisenberg resolution; K206 G3 honest tier-marked) - $CG_{\text{so5}} = \sqrt{(n_{\text{C}} - 1)} = 2$ at B_2 (standard so(n) trace formula; K206 G4) - FK norms $1/n_{\text{C}}$ and $2/(n_{\text{C}} \cdot C_2)$ (Heckman-Opdam Pochhammer Step 4) - $c_{\text{FK}} = 225/\pi^{(9/2)}$ (T2442 RATIFIED Saturday) - Wave functions $f_i(z) = z_i$ and $f_{ij}(z) = z_i z_j - z_j z_i$ (Step 3 explicit) - $z_{\text{source}} = \text{Shilov boundary general point } z_0$ (K206 G5) - Bergman intrinsic ℓ_{B} (Path δ closure)

No factor-fitting. No pattern-match shortcuts. Each step pinned from standard machinery (so(5) rep theory + FK Ch. XII-XIII + Heckman-Opdam).

Multi-week remaining (~1-2 weeks): - Step 6.2 P_op explicit FK action (~2 days) - Step 6.3 index sum + CG combination (~2 days) - Step 6.4 c_{FK} normalization absorption (~2 days) - Step 7 dimensional bridge + $\hbar_{\text{BST}}$ identification (~3 days) - Step 8 G_{observed} comparison (~2 days)

Closure target: $G_{\text{predicted}}$ vs G_{observed} at Tier 2 STRUCTURAL (10^{-4} to 10^{-2}) or Tier 1 EXACT via redshift anchor ($\sim 10^{-10}$ via GPS-clock).

5. K206 audit framework — 7 explicit gates

Gate	Status
G1 — Schur application correct	PASS preliminary (standard rep theory)
G2 — Heisenberg conjugacy justified	PASS (Lyra option (b): mass = K-type Casimir eigenvalue; $m_{\lambda} = \sqrt{C_2(\lambda)}$)
G3 — Substrate-coincidence tier-marked	OPERATIONAL (G3.a/b/c/d capture 4 Cal #35 firings)
G4 — Clebsch-Gordan multiplicity VERIFIED-CITED	Elie Step 2: $V_{-}(1,1) \otimes V_{-}(1,0) = 10 + 35 + 5$; multiplicity 1 $\square$
G5 — z_{source} canonicalization	Shilov boundary general point z_0
G6 — Matrix element computation closed form	PENDING Step 6.2-6.4 (~1 week)
G7 — Dimensional bridge $\rightarrow G_{\text{predicted}}$ SI	PENDING Step 7-8 (~5 days)

5/7 gates substantively documented. G6 + G7 pending Steps 6-8.

6. Tier 0 v0.2 consolidation — Session 2 active

Casey signal: GO. Lyra + Keeper joint multi-hour focused work begins.

Scope: Single coherent v0.2 draft absorbing: - v0.1 commitment operator + substrate time - v0.1.5 topology addendum (Reading A + dual bases; Shilov boundary commitment) - v0.1.5 $G_{\text{substrate}}$ sketch (walked back per K204-PARTIAL) - v0.1.6 native field equation + holographic boundary - Monday Heisenberg conjugacy justification - Monday matrix element framework Steps 1-5 - K200 + K204-PARTIAL + K206 audit framework integration - Cal #35 STANDING methodology integration

Lyra: consolidating draft side Keeper: bringing sphere reconciliation + K-audit anchor map + G chain Step 1-5 + Cal #35 integration

Output: v0.2 single coherent draft ready for Cal #191 K200/K206 gates cold-read.

7. Cumulative audit-chain — 16 symmetric events

Day	Events
Saturday	7 (3 Grace + 2 Lyra + Cal #182 on Keeper + Keeper self-catch on v0.6)
Sunday	5 (3 Grace + Lyra v0.6 + Keeper self-catch on K-invariance pre-stage candidates)
Monday	4 (factor-2 + parallel-framework + Pochhammer-mechanism + 0.924-arithmetic-walk-back)
Total	16 cumulative symmetric events

Pattern at maturity: Cal catches → Keeper synthesizes → Lyra+Elie acknowledge → Grace patches in-place. Monday morning produced framework closure WITHOUT cascading. Each brake produced sharper formulation, not slower output. Discipline-as-generator pattern operational at full strength.

8. L-axes honest disposition (carried forward from v0.10, Monday updates)

L-axis	Topic	Monday EOD disposition	Tier
L1	D_IV ⁵ uniqueness	T953 + Strong-Uniqueness	RATIFIED + CANDIDATE
L2	Shilov SO(4)	+ ρ -pinning C19	STRUCTURALLY VERIFIED
L3	SO(2) does many jobs	4 jobs unified	STRUCTURALLY VERIFIED
L4	Bergman H ² substrate Hilbert	Tier 0 v0.1.6 operator level	STRUCTURAL
L5	Absolute scale	CANDIDATE STRUCTURAL	CANDIDATE
L6	Time + Koons tick	READING (paused) T2405 + Tier 0 substrate time emergent	RATIFIED
L7	Cosmological cycle	Casey-named Interstasis	CANDIDATE
L8	Alpha tower	Paper #110 + Tier 0 operator	STRUCTURAL
L-G	Newton's G from substrate	Matrix element framework Steps 1-5 framework-complete; G_predicted closed form pending Steps 6-8 (~1-2 weeks)	GOAL ACTIVE; framework-complete

9. Honest scope — what closed Monday vs what didn't

What's substantively closed: - Cal #35 STANDING ratified (formal discipline-stack closure) - 4 Casey-named principle candidates NAMED (names usable; STANDING promotion pending Cal cold-read) - Matrix element framework framework-complete Steps 1-5 (operator-clean reduction) - K206 audit framework 5/7 gates documented - Mon-event-4 (0.924 arithmetic walk-back) honestly absorbed

What's NOT closed Monday: - P1 actually dispatched (Casey email action pending) - G chain numerical evaluation (multi-week Step 6-8) - Tier 0 v0.2 single coherent draft (Session 2 in progress) - K206 G6 + G7 (pending ma-

trix element value + dimensional bridge) - 4 named principles → STANDING (pending Cal cold-read + K-audit)
 - Candidate #14 vs Candidate #12 scope decision (Casey deferred) - L5 work (paused per Saturday consensus)

10. Cal queue status

#	Status
Cal #185 (P1 v0.7)	PASS — dispatch-ready
Cal #186 (Lane D L4 v0.2)	pending Lyra filing
Cal #187 (Lane E Dictionary)	COMPLETE Monday morning
Cal #188 (Lane C bulk-color #35 independence)	pending
Cal #189 (K204-PARTIAL + Casey-named candidates)	activates per Casey naming approval
Cal #190 (TBD G chain pattern-match disposition)	pending (likely subsumed by #192)
Cal #191 (Tier 0 v0.2 K200/K206 gates)	pending Lyra v0.2 draft
Cal #192 (matrix element framework G6+G7)	pending Elie Step 6
Cal #35 STANDING formal entry	filing Monday (Casey ratification approval)

11. Monday substantive program-level inflection — honest framing

Monday produced: - External engagement test going live (P1 ready for Casey dispatch) - Methodology stack maturation (Cal #35 STANDING; 6-element discipline stack closes) - Operator-level G chain framework closure (Casey directional ask “derive G from substrate” realized at framework level via shortest route) - Casey-named principles expansion (10 STANDING + 5 CANDIDATES → 10 STANDING + 5 CANDIDATES with 4 newly NAMED) - Tier 0 v0.2 consolidation underway (Sunday-morning operator framework gets single coherent draft)

This is the substantive Monday program inflection. Multi-week completion work (G chain Step 6-8, v0.2 draft, named principles STANDING promotion) continues at sustainable cadence.

— Keeper. HSL v0.11 filed Monday substantive program inflection. Supersedes v0.10. Standing for next surfacing checkpoint (Session 2 v0.2 prep convergence with Lyra).